BEHNAM YOUSEFI

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Education

Master of Science in Information Technology Engineering Sep 2016 – Sep 2018
Islamic Azad University, Iran

- GPA: 3.8 / 4 (18.63 / 20)
- **Master's Thesis:** Empirical Evaluation of Software Quality in Object-Oriented Java Systems Using Static Analysis

Bachelor of Science in Information Technology Engineering Sep 2012 – Jan 2016
Islamic Azad University, Iran

- GPA: 2.9 / 4 (14.30 / 20)

Research Interests

- Software Engineering
- Machine Learning
- Data Science and Analysis
- Internet of Things
- Cybersecurity
- Optimization

Publications

- Yousefi, B., Taghipour, S., Toghra, M., *Empirical Evaluation of Object-Oriented Software Metrics in Java Systems*, Manuscript in preparation.
- Yousefi, B., Moosavi, M., Toghra, M., *A Static Analysis Framework for Software Quality Assessment in Java Applications*, Manuscript in preparation.

Professional Experience

IT Engineer 2012 – Present
Telecommunication Company of Iran (TCI)

- Maintained enterprise-scale telecommunication IT infrastructure and network services.
- Monitored network performance and analyzed operational logs to identify recurring faults and improve service reliability.
- Performed root-cause analysis of hardware, software, and network-related incidents.
- Automated routine monitoring and reporting tasks using Python and shell scripting.
- Supported system deployment and network service optimization.

Academic Experience

Research Assistant (RA) 2017 – 2018
Networking and Telecommunications Laboratory
Islamic Azad University

- Developed modules for a Java-based static analysis framework for object-oriented software systems.
- Implemented software quality metrics including CBO, LCOM, RFC, DIT, and WMC.
- Conducted empirical evaluation of software quality and maintainability across open-source Java projects.

Teaching Assistant (TA) Islamic Azad University

- **Object-Oriented Programming** | Dr. Morteza Mousavi — Sep 2016 – Jan 2017
- **System Analysis and Design** | Dr. Sara Taghipour — Feb 2017 – Jun 2017
- **Software Design and Architecture** | Dr. Mohammad Toghra — Sep 2017 – Jan 2018
- **Design and Analysis of Algorithms** | Dr. Morteza Mousavi — Sep 2017 – Jan 2018
- **Artificial Intelligence** | Dr. Mohammad Toghra — Feb 2018 – Jun 2018
- **Computer Networks** | Dr. Sara Taghipour — Feb 2018 – Jun 2018
- **Information Technology Fundamentals** | Dr. Mohammad Toghra — Feb 2018 – Jun 2018

Selected Projects

- **Static Analysis Framework for Java Software Quality Assessment**
  - Developed a Java-based static analysis tool to extract structural information from object-oriented software systems.
  - Implemented object-oriented software metrics including CBO, LCOM, RFC, DIT, and WMC.
  - Evaluated software quality and maintainability using empirical analysis.
  - Applied the framework to multiple open-source Java projects.
- **IoT Service Anomaly Detection Using Multisource Log Fusion**
  - Investigated anomaly detection techniques for IoT services using multisource log fusion.
  - Integrated and analyzed call, business, and metric logs to identify fault propagation across enterprise services.
  - Applied statistical and graph-based methods to detect anomalies in IoT service performance.
  - Evaluated detection performance using accuracy, precision, recall, and F1-score.
- **Network Performance Monitoring and Fault Analysis**
  - Collected and analyzed operational logs from enterprise network infrastructure.
  - Investigated network performance indicators to identify recurring faults and improve service reliability.
- **Hybrid Intrusion Detection Model**
  - Engineered statistical and flow-based network traffic features for attack detection.
  - Developed a PSO-optimized SVM classifier and compared it with GA-based feature selection.
  - Evaluated detection performance across multiple attack scenarios.

Skills & Tools

- **Programming:** Java – Python – MATLAB – SQL – JavaScript
- **Data Analysis & Machine Learning:** NumPy – Pandas – Scikit-learn – Statistical Analysis
- **Networking & Systems:** TCP/IP – Network Monitoring – Fault Diagnosis – Enterprise IT Infrastructure – Internet of Things (IoT)
- **Tools:** Git – Eclipse – IntelliJ IDEA – Jupyter Notebook – LaTeX
- **Operating Systems & Scripting:** Linux – Windows – Bash

Languages

- **Persian:** Native
- **English:** TOEFL — June 2026
  Overall: 4.5/6 (Reading: 6.0/6 , Listening: 5.0/6 , Writing: 4.5/6 , Speaking: 3.0/6)

Certifications

- Machine Learning – Sharif University of Technology
- Data Analysis with Python
- IoT Fundamentals – Cisco Networking Academy
- Software Design and Architecture – Coursera
- CompTIA Network+ Training
- Mikrotik Routing and Firewall Training

Honors & Awards

- Ranked 1st among M.Sc. students in Information Technology Engineering at Islamic Azad University.
- Received the Academic Excellence Scholarship for outstanding graduate performance.
- Received the Distinguished Master's Thesis Award for research on software quality assessment using static analysis.